Week 01 - Tutorial 05

Transcript Language

English

Application of vector addition and scalar multiplication: working with some large data

set

Hello.

So in this video we will see an application of vector addition and scalar multiplication.

So where we will consider some large dataset and we will how we can apply vector addition

and scalar multiplication which you have learned in this Week 1.

So let us take this dataset where in the left hand side we have airlines and in the right

hand side we have safety scores.

These safety scores is basically an indicator whether the airline is safe or not.

So it depends on the number of accidents and number of fatalities occurred in those accidents.

And so where in column one we are taking time period from 1985 to 1999 and in the second

column we are taking the time period from 2000 to 2014.

And in the third column we are seeing the combined.

So let us denote this as C3.

So in column one there are some entries V1, V2, V3 and so on and in column two there are

some entries like W1, W2, W3 and so on.

So when we are adding C1 and C2, we are basically adding coordinate wise that means V1 plus

W1, V2 plus W2, V3 plus W3 and so on, which we have seen in the lecture.

And what C3 represents, C3 represent is the average of C1 and C2.

So C3 is nothing but half of V1 plus W1 will be in the first row, half of V2 plus W2 will

be in the second row.

So this is the scalar multiplication.

This is the scalar.

And we are multiplying with the entry in the first row which is V1 plus W1.

This is the scalar half and we are multiplying it with the vector, with the entry V2 plus

W2 which is in the second row.

That is how we are computing C3.

It is difficult to do it by hand.

So what we are doing, so we can use this excel sheet to find the entries.

So this column is denoted as B here in the excel sheet and this column is denoted as

C in the excel sheet.

So what we have done here, you can see that we are taking B3, sum of B3 and C3.

B3 is this one because this is the third row.

First two row are used for this naming of this column.

So we are starting from the third row.

So we are taking sum of B3 and C3 and we write it here and similarly this will be sum of

B4 and C4, this will be sum of B5 and C5 and so on.

And then if we just scalar, the if we just multiply these entries with half, which is

a scalar in real number basically, then we will get this column, the combined column.

So that is how you can see that using this vector addition which eventually we will use

in programming then it will be easier to work with this large dataset or even you can use

this excel sheet what I am using here to calculate the sum of two vectors, large vectors basically

or calculating the scalar multiplication of those large vector.

Thank you.